

JAKUB KIERZNOWSKI

Junior Software Engineer | Software Developer Intern at IBM
CS & Intelligent Systems Student at AGH

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SUMMARY

Final-year Computer Science student at AGH (GPA 4.74/5.00, graduating January 2027) and Software Developer Intern at IBM since February 2026, working on Data Lineage in Java and Spring. I built and run InstalDesk, an invoicing app for Polish installers that files invoices straight to KSeF. I have also built LLM agents on Neo4j and trained multi-agent RL policies, and my thesis compares 3D reconstruction models. **Looking for a junior software engineer role where product engineering meets applied AI.**

Java 21 · Spring Boot · Spring Security · PostgreSQL · Neo4j · Python · FastAPI · Docker · RabbitMQ and MQTT · Jenkins · GitHub Actions

EXPERIENCE

Software Developer Intern

IBM Software Lab · Kraków · hybrid
Feb 2026 – Present

- Optimise graph and tree traversal in Java/Spring; extend Data Lineage APIs with graph-derived and enriched metadata.
- Implemented production Neo4j label migrations in transactional batches and added full-text indexes for Lucene-based search.
- Write unit and API tests with JUnit, Mockito and pytest in Jenkins CI/CD; PRs must meet a 90% SonarQube coverage gate.
- Reduced pytest BVT suite runtime from 20 minutes to 4 through asynchronous execution, parametrised cases and session-scoped fixtures.

IT Team Lead (Web & Workflow Automation)

Student Council · WRSS WEAlilB, AGH
Mar 2025 – Present

- Maintain the council website (Vue, Tailwind CSS) and the VPS it runs on.
- Built internal automations in self-hosted n8n for Slack and Google Workspace (OAuth 2.0).
- Coordinated the VOLT 2025 tech conference and its deals with eight partner companies, including Aptiv, PwC Polska and Cytiva.

Team Mentor and Community Leader

Zwolnieni z Teorii · remote
Sep 2022 – May 2025

Mentored 35+ student teams per edition on social-impact projects. Later ran the participant community and did olympiad user support.

Game Developer

AGH Code Industry · Kraków
Dec 2023 – Jun 2024

Gameplay mechanics in Unity and C# for “Student Trainer”, a game made by the student club.

PROJECTS

Code at github.com/qualv13

InstalDesk, invoicing SaaS for Polish installation contractors

instaldesk.pl · live since Apr 2026, pilot users · closed source · Python, FastAPI, SQLAlchemy, SQLite, Stripe, KSeF API

- Sends invoices directly to KSeF, the Polish government e-invoicing system (API v2, FA(3) schema), with no third-party integrator. Handles receipts, offline mode and importing cost invoices.
- Web app (PWA) and a Windows desktop client on one account. The desktop client keeps a local SQLite database, so it works offline.
- Stripe subscriptions (card, BLIK, Przelewy24), two-factor login, user roles, tenant isolation and encrypted backups.
- 1,400+ unit and integration tests in nightly CI. Full-stack and solo: I build it, deploy it and handle production support.

IoT fleet backend

IoTServerApp · Java 17, Spring Boot 3.3, PostgreSQL, RabbitMQ/MQTT, Protocol Buffers, Docker

Spring Boot backend for managing fleets of smart lamps. Devices talk to it over MQTT (RabbitMQ) using Protocol Buffers. JWT login with refresh tokens, scheduled device jobs and OTA firmware updates from S3. Runs with Docker Compose.

DrugPath, LLM agent over a biomedical knowledge graph

neo4j-agent · Neo4j Aura Agent Hackathon 2026, top 6 · Neo4j Aura, Cypher, Python, LLM agents

Walks the Hetionet graph (47k nodes, 293k relationships) from drug to gene, pathway and disease to explain how a drug works and where it could be repurposed. Four tools: three Cypher queries and a vector search. Demo: qualv13.github.io/neo4j-agent

Hardware Hub, inventory and rental system

hardware-hub · recruitment assignment · Python, FastAPI, Neo4j, Vue 3, Gemini, Docker

Natural-language inventory search with Gemini tool calling, including requests for faulty devices. Computes compatibility by traversing a Neo4j graph (device → port → machine → OS) and shows the connection path. Includes a non-LLM fallback, 72 pytest tests in CI and a fixed query set comparing graph-based and plain search.

Image-to-3D reconstruction, engineering thesis

AGH thesis · 2026 · Python, PyTorch, FastAPI

Trained my own occupancy network (ResNet-18 encoder, FiLM decoder, 15M parameters) that rebuilds a 3D shape from one to three photos. Benchmarked it against TripoSR, InstantMesh, TRELIS and TripoSG (0.4–2.3B parameters) on ShapeNet, Google Scanned Objects and Pix3D, with paired significance tests.

Multi-agent reinforcement learning for a local energy market

research project · AGH, paper in preparation · Unity ML-Agents, C#, PyTorch, PPO, ONNX

32 households with solar panels and batteries trade energy with each other in a Unity simulation. Two-person project: I prepared the solar and consumption data (PVGIS, OPSD), built the environment and trained the agents with PPO, then compared them with DQN, A2C and rule-based agents across several seeds.

Graph and deep neural networks

GGSN · AGH coursework · PyTorch, TensorFlow, scikit-learn

Compared four graph neural networks (GCN, GAT, GraphSAGE, GIN) on the Yelp dataset. Also transfer learning with Grad-CAM, DistilBERT sentiment classification and model compression.

HACKATHONS AND COMPETITIONS

- **Wincent DragonByte 2026, Bratislava:** competed in the onsite final of this algorithmic programming contest on a wild card.
- **Google Student AI Hackathon 2026, Warsaw:** built ShowMeTheWay in one day with three teammates. It reads a CV and a goal, plans a learning roadmap and puts the steps into Google Calendar (Gemini API).
- **HackYeah 2025:** top 8 of 65+ projects in the biohacking track with Fit2Beat (fit2beat.toadres.pl), which builds workout playlists from a target BPM. Team of four, 24 hours.

SKILLS

Core: Java 21, Spring Boot, Spring Security, Hibernate/JPA, Maven, Python, FastAPI, SQL, PostgreSQL, Neo4j, Cypher, REST APIs, Git, unit and integration tests (JUnit 5, Mockito, pytest), OOP and design patterns, data structures and algorithms.

AI: LLM agents and tool calling (Gemini API, Neo4j Aura Agents), PyTorch, reinforcement learning (PPO), graph neural networks, TensorFlow, scikit-learn, Ollama.

Working knowledge: Docker and Compose, Linux and Bash scripting, CI/CD (Jenkins, GitHub Actions), RabbitMQ, MQTT, Protocol Buffers, SQLAlchemy, SQLite, Testcontainers, OAuth 2.0, AWS S3.

Also used: Vue, React, TypeScript, Tailwind CSS, C# (Unity), C++, n8n.

Process: Agile (Scrum, Kanban), GitHub Projects, Jira, AI-assisted development (Claude Code, Codex, IBM Bob).

Languages: Polish (native), English (C1), French (A2).

CERTIFICATIONS

Neo4j Certified Professional and Building Agents in Neo4j Aura, both 2026.

Amazon Junior Software Developer (Coursera specialisation): Data Structures and Algorithms, Programming with Java, Application Development.

INTERESTS

Snowboarding, sailing, board games, game dev, and a home server running Docker and local LLMs.

EDUCATION

Computer Science and Intelligent Systems, BEng

AGH University of Krakow · Kraków

Oct 2023 – Jan 2027

GPA 4.74/5.00. Year Representative since the 1st semester.

Extended Mathematics, Physics and Computer Science

V High School · Kraków

Sep 2019 – Jun 2023

C++ and competitive programming.